

TANMAY MANGAL

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science and Engineering — CGPA: **8.49 / 10.0**

Bhopal, Madhya Pradesh

Sep 2024 – May 2028

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks and Tools: React.js, Next.js, FastAPI, Tailwind CSS, Pandas, NumPy, Git, GitHub, Vercel, SQLite

Concepts: Full-Stack Development, Algorithmic Trading, REST APIs, OOP, Data Structures and Algorithms

EXPERIENCE

Quantitative Research Intern

Remote

TradeVed Fintech Labs Private Limited

Jun 2026 – Jul 2026

- Contributing to the development of TradeVed's stock screener features, including filter logic, data validation, and UI workflows for retail-trader use cases
- Backtesting systematic trading strategies on historical NSE data using Python and Pandas to evaluate indicator performance and risk-reward characteristics
- Collaborating with the product team to translate quantitative research into production-ready features for the TradeVed fintech platform

Web Developer Intern

Remote

Dynamic Bubble, Digital Marketing Agency

May 2026 – Present

- Engineered and deployed the company's full production website (dynamicbubble.agency) from scratch using React.js, Node.js, and MongoDB
- Maintained and scaled the live site by regularly shipping new features, UI improvements, and performance optimizations
- Collaborated with the founder to translate branding requirements into polished, production-ready web solutions

AI Training Data Contributor, ML Engineer

Remote

Shipd by Datacurve (YC W24)

May 2026 – Present

- Designed and developed 10+ coding challenges that stump state-of-the-art foundation models, contributing high-quality data to Shipd's AI training pipeline
- Engineered problem sets to strengthen large language model reasoning and code generation capabilities on Datacurve's platform
- Selected as a contributor on Shipd (YC W24), a platform that has paid out \$2M+ to problem solvers for the world's top AI labs

PROJECTS

MAET: Market Analytics and Execution Terminal | *Next.js, TypeScript, Python, FastAPI, WebSocket* [Live](#) | [GitHub](#)

- Architected a real-time NSE algorithmic trading terminal with WebSocket tick streaming across a 7-module workspace: trade, markets, charts, portfolio, strategy, risk, and journal
- Integrated 5 technical indicators (VWAP, EMA, RSI, MACD, Bollinger Bands), candlestick pattern detection, and paper trading mode replicating institutional trading workflows
- Implemented 600+ passing tests, backend risk checks, and admin validation to support controlled paper-trading with live order execution disabled for safety

FinCalc India | *Next.js, TypeScript, PostgreSQL, Prisma ORM, NextAuth*

[Live](#) | [GitHub](#)

- Architected a suite of 6 real-time financial calculators covering SIP, EMI, FD, PPF, Lumpsum, and Income Tax FY 2024-25 with Old vs New regime comparison
- Built REST APIs with schema validation and Google OAuth; deployed interactive charts on Vercel with an automatically provisioned PostgreSQL database
- Designed responsive input forms with validation flows across 6 calculator workflows, improving user completion and accuracy

Lumint: AI Fraud Intelligence Platform | *Python, FastAPI, Next.js, TypeScript, SQLite, OpenCV*

[Live](#) | [GitHub](#)

- Engineered a full-stack AI fraud detection platform with 4 detection modules: document forensics, phishing URL analysis, UPI payment screenshot review, and fraud campaign clustering
- Leveraged Groq's LLaMA 3.3 70B to generate natural-language fraud explanations and built a cross-modal risk fusion engine that combines all 4 module signals into one unified threat score
- Implemented OCR-based extraction, URL risk scoring, QR parsing, and confidence summaries across 11 frontend routes for analyst-style risk review

CERTIFICATIONS

[Trading Algorithms](#) | Indian School of Business (ISB) via Coursera | Credential ID: **SI5TYMHGUUSD**